Analyst / Technical Report — AMF-WIN-SAMPLE001

Report ID: `4159bc46eddc9b8adc32dbc84db51554b9e72524d0b44f63d2206affe235aa16` · *Snapshot:* 2026-06-07T01:30:33.897125+00:00

> **Likelihood scale (FIRST 5-tier):** almost_certain > highly_likely > likely > unlikely > remote. Likelihood estimates the probability of the assessed activity; it is kept separate from analytic confidence.
>
> **Confidence (LCA):** high / moderate / low — the analyst's confidence in the assessment given evidence quality and corroboration. Distinct from likelihood.

Findings

malfind recovered 15 executable VAD hits (11 PAGE_EXECUTE_READWRITE / RWX, 4 PAGE_EXECUTE_READ / RX) across 5 processes; RWX concentrated in winlogon.exe (PID 628, x9)

- **Finding ID:** `F-AMF-S001-001` · **Severity:** medium · **Likelihood:** unlikely · **Confidence:** moderate · **Risk score:** 6
- **MITRE ATT&CK:** `T1055`

get_malfind (volatility3.windows.malfind.Malfind) reported 15 executable VAD hits: 11 RWX (PAGE_EXECUTE_READWRITE) and 4 RX (PAGE_EXECUTE_READ). RWX by process: winlogon.exe PID 628 x9, csrss.exe PID 604 x1, msimn.exe PID 1984 x1. RX (not RWX): winlogon.exe PID 628 x1 (0x580000), lsass.exe PID 692 x2, msmsgs.exe PID 548 x1. RWX is concentrated in winlogon.exe, the canonical AMF process-injection teaching signal.

Evidence:
```
malfind RWX winlogon.exe PID 628 x9
(0x42e20000,0x22f40000,0x24c00000,0x248a0000,0x40a60000,0x2b7f0000,0x434000
vad_tag VadS PAGE_EXECUTE_READWRITE
malfind RWX csrss.exe PID 604 0x7f6f0000 vad_tag Vad
PAGE_EXECUTE_READWRITE
malfind RWX msimn.exe PID 1984 0x1eb0000 vad_tag VadS
PAGE_EXECUTE_READWRITE
malfind RX winlogon.exe PID 628 0x580000 vad_tag Vad
```

```
PAGE_EXECUTE_READ
malfind RX lsass.exe PID 692 0x280000,0x7f6f0000 vad_tag Vad
PAGE_EXECUTE_READ
malfind RX msmsgs.exe PID 548 0x520000 vad_tag Vad
PAGE_EXECUTE_READ
plugin volatility3.windows.malfind.Malfind hit_count=15
```

9 RWX (PAGE_EXECUTE_READWRITE) malfind regions in winlogon.exe (PID 628, ppid 356) — dominant injected/unpacked-code concentration

- **Finding ID:** `F-AMF-S001-002` · **Severity:** medium · **Likelihood:** unlikely · **Confidence:** moderate · **Risk score:** 6
- **MITRE ATT&CK:** T1055

get_malfind reported 10 executable VAD hits in winlogon.exe (PID 628, ppid 356 smss.exe), of which 9 are RWX (PAGE_EXECUTE_READWRITE, vad_tag VadS) and 1 is RX (PAGE_EXECUTE_READ at 0x580000, vad_tag Vad). The 9 RWX regions are the dominant injected/unpacked-code concentration in this image and the canonical AMF process-injection teaching signal.

Evidence: `winlogon.exe PID 628 RWX VadS:`
```
0x42e20000,0x22f40000,0x24c00000,0x248a0000,0x40a60000,0x2b7f0000,0x434000
(9 x PAGE_EXECUTE_READWRITE)
winlogon.exe PID 628 RX Vad: 0x580000 (PAGE_EXECUTE_READ, not RWX)
pslist: winlogon.exe pid 628 ppid 356
plugin volatility3.windows.malfind.Malfind
```

2 PAGE_EXECUTE_READ (RX, not RWX) malfind regions in lsass.exe (PID 692) — read-execute, no write permission

- **Finding ID:** `F-AMF-S001-003` · **Severity:** low · **Likelihood:** unlikely · **Confidence:** moderate · **Risk score:** 4
- **MITRE ATT&CK:** T1055

get_malfind reported 2 executable VAD regions in lsass.exe (PID 692, ppid 628 winlogon.exe), both PAGE_EXECUTE_READ (RX), vad_tag Vad — NOT PAGE_EXECUTE_READWRITE. An earlier draft mislabeled these as RWX injection into the credential process; the authoritative malfind output (step6_malfind_300s.json) shows protection PAGE_EXECUTE_READ for both, so these are read-execute mapped regions with no write permission and are not RWX injection. Lower confidence accordingly.

Evidence: `lsass.exe PID 692 0x280000 vad_tag Vad PAGE_EXECUTE_READ (RX)`
```
lsass.exe PID 692 0x7f6f0000 vad_tag Vad PAGE_EXECUTE_READ (RX)
pslist: lsass.exe pid 692 ppid 628
plugin volatility3.windows.malfind.Malfind
```

1 RWX (PAGE_EXECUTE_READWRITE) malfind region in csrss.exe (PID 604)

- **Finding ID:** `F-AMF-S001-004` · **Severity:** medium · **Likelihood:** unlikely · **Confidence:** moderate · **Risk score:** 6
- **MITRE ATT&CK:** `T1055`

get_malfind reported 1 RWX VAD region in csrss.exe (PID 604, ppid 356 smss.exe) at 0x7f6f0000, vad_tag Vad, PAGE_EXECUTE_READWRITE.

Evidence: `csrss.exe PID 604 0x7f6f0000 vad_tag Vad PAGE_EXECUTE_READWRITE (RWX)`
```
pslist: csrss.exe pid 604 ppid 356
plugin volatility3.windows.malfind.Malfind
```

1 RWX (PAGE_EXECUTE_READWRITE) malfind region in msimn.exe (PID 1984); msmsgs.exe (PID 548) region is PAGE_EXECUTE_READ (RX, not RWX)

- **Finding ID:** `F-AMF-S001-005` · **Severity:** low · **Likelihood:** unlikely · **Confidence:** moderate · **Risk score:** 4
- **MITRE ATT&CK:** `T1055`

get_malfind reported 1 RWX VAD region in msimn.exe (Outlook Express, PID 1984) at 0x1eb0000, vad_tag VadS, PAGE_EXECUTE_READWRITE. The msmsgs.exe (Windows Messenger, PID 548) region at 0x520000 is PAGE_EXECUTE_READ (RX), vad_tag Vad — NOT RWX. An earlier draft labeled both as RWX; the authoritative malfind output shows only msimn.exe is RWX. RWX in these GUI apps is commonly benign JIT/packing on XP-era images, hence low confidence.

Evidence: `msimn.exe PID 1984 0x1eb0000 vad_tag VadS PAGE_EXECUTE_READWRITE (RWX)`
```
msmsgs.exe PID 548 0x520000 vad_tag Vad PAGE_EXECUTE_READ (RX, not RWX)
plugin volatility3.windows.malfind.Malfind
```

Process inventory recovered: 21 running processes, coherent PPID forest (2 roots, 0 orphans, 0 LOLBin flags)

- **Finding ID:** `F-AMF-S001-006` · **Severity:** low · **Likelihood:** unlikely · **Confidence:** moderate · **Risk score:** 4
- **MITRE ATT&CK:** `T1057`

get_pslist returned 21 processes (System pid4, smss 356, csrss 604, winlogon 628, services 680, lsass 692, svchost 852...); build_process_tree: process_count 21, root_count 2 (System, smss.exe), orphan_count 0, suspicious_count 0. cmdline recovered all 21 command lines consistent with pslist. get_netscan recovered 0 sockets (honest empty real result on this XP-era image). Baseline process-discovery context for the injection findings.

Evidence: `get_pslist process_count=21`
```
build_process_tree: process_count 21, root_count 2 (System pid4, smss.exe pid356), orphan_count 0, suspicious_count 0
get_netscan socket_count=0
get_svcscan service_count=229
cmdline rows=21
```

Indicators of Compromise

No IOCs extracted.

Timeline

(provenance graph omitted in PDF — see the IOC table above; renders in the HTML/GitLab view)

Timestamp	Host	Event	Phase	Description
2026-06-06T20:36:37Z	sample001	TL-AMF-S001-001	—	case_init: AMF-WIN-SAMPLE001 created victor.galvan, severity medium, scope /c windows/sample001.bin
2026-06-06T20:48:49Z	sample001	TL-AMF-S001-002	—	evidence_register: sample001.bin sha25 03242077eb3364fb248d1c7730fd0a940 size 536330240 bytes, indexed to agent
2026-06-06T20:50:05Z	sample001	TL-AMF-S001-004	—	get_pslist 21 processes; get_netscan 0 s image); get_svcscan 229 services; build_ orphans, 0 suspicious)
2026-06-06T23:17:43Z	sample001	TL-AMF-S001-005	—	approve_finding DRAFT->APPROVED for victor.galvan, approval_id 4a881577139b59efadb980816d47adfce NOTE: Playwright-automated demo app
2026-06-06T20:50:00Z	sample001	TL-AMF-S001-003	—	get_malfind (volatility3.windows.malfind executable VAD hits — 11 PAGE_EXECUT PAGE_EXECUTE_READ (RX). RWX by proc csrss.exe x1 (PID 604), msimn.exe x1 (PI x1 (PID 628, 0x580000), lsass.exe x2 (PID